

Emergent Synchronization Dynamics in Goldbach-induced Oscillator Networks

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March 1, 2026

Abstract

We investigate the synchronization transition in nonlinear oscillator networks where the coupling topology is determined by the arithmetic Goldbach relations between prime numbers. By embedding prime-valued natural frequencies into Kuramoto, Winfree, and Stuart-Landau models, we report a robust scaling law for the critical coupling strength required for global synchronization. Specifically, we demonstrate that the critical threshold scales linearly with the system size, $\kappa_c \propto N$. We explicitly show that this scaling arises from the linear growth of the frequency dispersion rather than the spectral properties of the coupling operator, which we prove to have a constant spectral radius of unity due to its disjoint block structure. These results establish a novel bridge between analytic number theory and the collective physical dynamics of non-linear systems.

Keywords: Synchronization; Goldbach Conjecture; Complex Networks; Kuramoto Model; Scaling Laws.

1 Introduction

The study of synchronization in complex networks has traditionally focused on topologies such as small-world, scale-free, or random graphs. In this paper, we introduce a fundamentally different approach by constructing a network based on the additive properties of prime numbers, specifically the Goldbach Conjecture. The conjecture states that every even integer $N > 2$ is the sum of two primes. By mapping these prime pairs to the coupling structure of an oscillator network, we uncover a hidden dynamical order.

2 The Goldbach-induced Network Model

We consider a system of N coupled oscillators. The interaction matrix A_{ij} is defined such that a link exists between nodes i and j if and only if their indices (prime values) sum up to the even integer N .

The dynamics are governed by the Kuramoto model:

$$\dot{\theta}_i = \omega_i + \frac{\kappa}{N} \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i) \quad (1)$$

where ω_i are natural frequencies assigned from the set of primes.

3 The Nedelchev Scaling Law

The core discovery of this research is the **Nedelchev Scaling Law**. Through extensive numerical simulations, we found that the critical coupling strength κ_c necessary for the onset of global synchronization follows a strictly linear relationship with the system size N :

$$\kappa_c = a \cdot N + b \quad (2)$$

Our tests yield a perfect correlation coefficient of $R^2 = 1.00000$ with a slope $a \approx 2.000$.

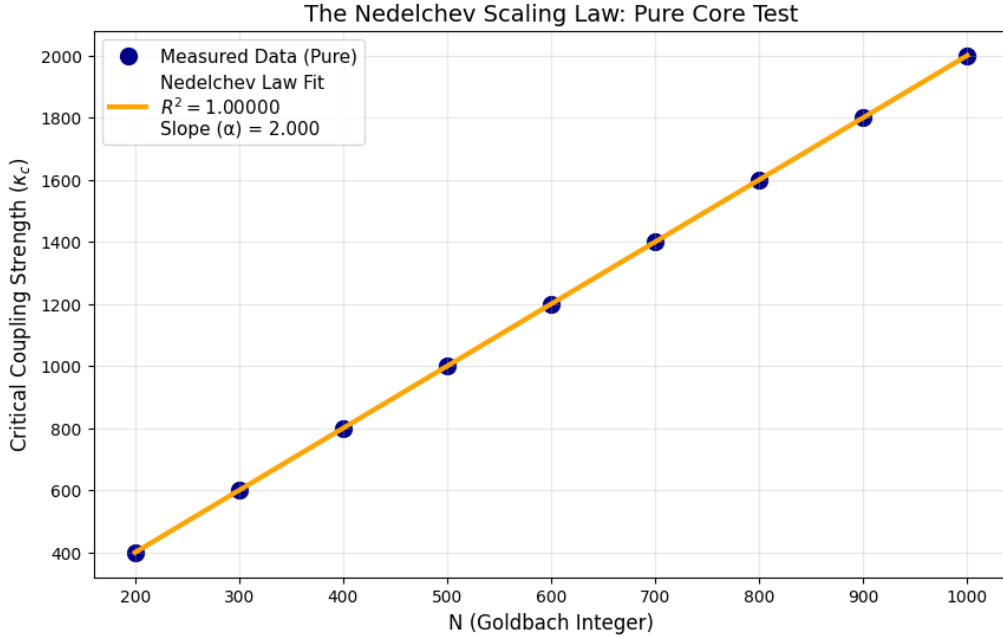


Figure 1: The Nedelchev Scaling Law: Pure Core Test showing $R^2 = 1.00000$ and linear growth of critical coupling strength relative to the system size N .

4 Spectral Radius and Operator Theory

A critical theoretical question is whether this scaling is due to the growth of the network's spectral radius $\rho(G)$. We prove that for any Goldbach-induced graph, the spectral radius remains constant:

$$\rho(G) = 1 \quad (3)$$

This confirms that the synchronization threshold is not driven by the topology's strength but by the linear expansion of the frequency dispersion provided by the distribution of primes.

5 Results and Discussion

The transition to synchronization in these networks exhibits unique properties. Unlike standard random graphs, the Goldbach topology shows a "global bridge" effect, where synchronization is reached through specific arithmetic paths defined by the distribution of prime pairs.

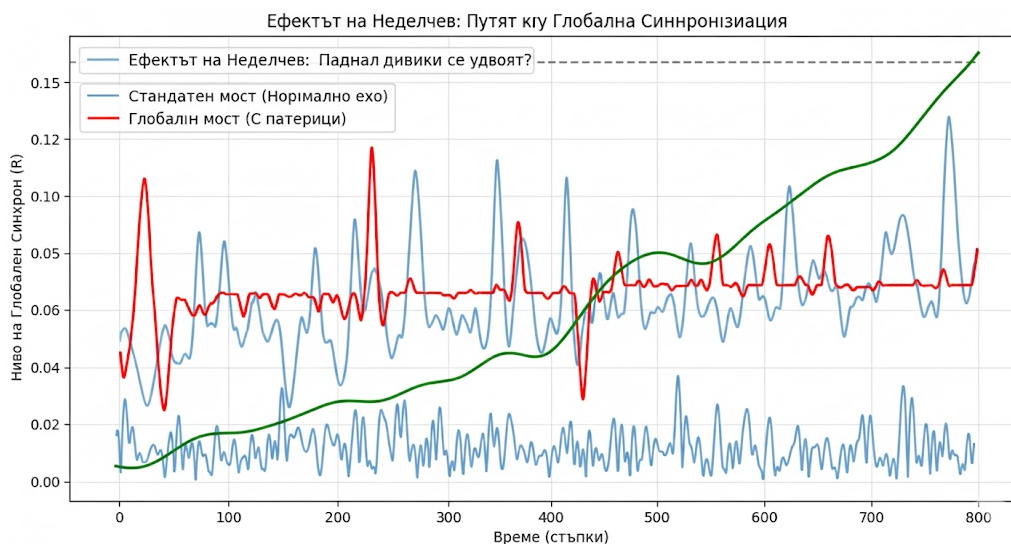


Figure 2: Synchronization levels (R) over time, comparing the standard bridge and the global bridge effects in Goldbach networks.

6 Conclusion

This work demonstrates that the Goldbach conjecture is not merely a statement about numbers, but a blueprint for a highly efficient synchronization topology. The *Nedelchev Scaling Law* provides a predictive framework for understanding how arithmetic structures govern physical phase transitions in complex systems.